



Homework 3

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Problem 3.1

Proof.

$$S(n) \equiv \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad \forall n \in \mathbb{N}, n \geq 1$$

Let us define $C(n)$ as

$$C(n) = \left\{ n \in \mathbb{N} \mid \sum_{k=1}^n k^2 \neq \frac{n(n+1)(2n+1)}{6} \right\}$$

We assume $C(n)$ is not null so there should exist a least element n .

We check $n = 1$

$$\left\{ \begin{array}{l} \sum_{k=1}^1 k^2 = 1^2 = 1 \\ \frac{1(1+1)(2 \cdot 1 + 1)}{6} = \frac{6}{6} = 1 \end{array} \right\} \implies 1 \notin C(n) \implies n > 1$$

There should exist a least element $n > 1$ in $C(n)$ so $n - 1 \notin C(n)$

We prove that for some $n \in \mathbb{N}^*$ the $S(n - 1) + n^2 = S(n)$ does not hold

$$\left\{ \begin{array}{l} S(n - 1) + n = \sum_{k=1}^{n-1} k^2 = \frac{(n-1)n(2(n-1)+1)}{6} + n = \frac{2n^3 - 3n^2 + n}{6} + n = \frac{2n^3 + 3n^2 + n}{6} \\ S(n) = \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} = \frac{2n^3 + 3n^2 + n}{6} \end{array} \right\} \implies$$

$$\implies S(n - 1) + n^2 = S(n)$$

So $C = \emptyset$ and $S(n)$ holds for all $n \in \mathbb{N}, n \geq 1$ □

Problem 3.2

- a) *Proof.* Lets us define a set $S = \{5a + 8b \mid a, b \in \mathbb{N}\}$ we assume 12\$ is in S and can be made from 5 and 8\$ gift cards then we can write 12 as $5a + 8b = 12$ and we check each possible combination:

$12 - 8 = 4$, because $4 < 8$ and $4 < 5$ so we can't make 12\$ from one 8\$ card

$12 - 5 = 7$, because $7 < 8$ we are left with only 5 and $7 - 5 = 2$ so it can't be made from 2 5\$ cards. And because the sum can't be negative and we proved that none of the possible combinations of 5 and 8\$ cards hold we proven that we can't make 12\$ from 5\$ and 8\$ cards □

- b) *Proof.* Suppose, for contradiction, that some integer $n \geq 28$ is not formable. Then the set

$$C = \{n \in \mathbb{N} : n \geq 28 \text{ and } n \text{ is not formable}\}$$

is a nonempty set of natural numbers. By the Well Ordering Principle, C has a least element m . Since $m \geq 28$, we have two cases.

Case 1: $28 \leq m \leq 32$. All elements are formable:

$$28 = 4 \cdot 5 + 1 \cdot 8,$$

$$29 = 1 \cdot 5 + 3 \cdot 8,$$

$$30 = 6 \cdot 5 + 0 \cdot 8,$$

$$31 = 3 \cdot 5 + 2 \cdot 8,$$

$$32 = 0 \cdot 5 + 4 \cdot 8.$$



Case 2: $m \geq 33$. Then $m - 5 \geq 28$. We can write $m - 5 = 5a + 8b$ $a, b \geq 0$.
Then

$$m = 5(a + 1) + 8b,$$

so m is formable, contradicting $m \in C$.

Both cases lead to a contradiction, so C is empty. Therefore every amount $n \geq 28$ can be formed using \$5 and \$8 cards. $\square$

Problem 3.3

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots + \frac{1}{n^2} < 2 - \frac{1}{n}, \quad \forall n \in \mathbb{N}, n > 1.$$

Proof. Lets us define: $f(n) = \sum_{i=1}^n \frac{1}{i^2}$.

1. Base case ($n = 2$): For $n = 2$, if we evaluate both sides we get:

$$\text{LHS} = f(2) = 1 + \frac{1}{4} = \frac{5}{4} = 1.25, \quad \text{RHS} = 2 - \frac{1}{2} = \frac{3}{2} = 1.5.$$

Since $1.25 < 1.5$, the inequality holds true.

2. We prove the induction

$$1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{k^2} < 2 - \frac{1}{k}.$$

$$1 + \frac{1}{4} + \cdots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k} + \frac{1}{(k+1)^2}.$$

$$2 - \frac{1}{k} + \frac{1}{(k+1)^2} \leq 2 - \frac{1}{k+1} \iff \frac{1}{(k+1)^2} \leq \frac{1}{k} - \frac{1}{k+1}.$$

$$\frac{1}{k} - \frac{1}{k+1} = \frac{(k+1) - k}{k(k+1)} = \frac{1}{k(k+1)}.$$

$$\frac{1}{(k+1)^2} < \frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

$$1 + \frac{1}{4} + \cdots + \frac{1}{(k+1)^2} < 2 - \frac{1}{k} + \left(\frac{1}{k} - \frac{1}{k+1} \right) = 2 - \frac{1}{k+1}.$$

$\square$



Problem 3.4

$$F_1 + F_2 + F_3 + \cdots + F_n = F_{n+2} - 1, \quad \forall n \in \mathbb{N}, n \geq 1.$$

Proof. From the definition of the Fibonacci sequence: $F_1 = 1$, $F_2 = 1$, and $F_k = F_{k-1} + F_{k-2}$ for all integers $k \geq 3$.

We define be the statement: $f(n) = \sum_{i=1}^n F_i = F_{n+2} - 1$.

1. Base Case ($n = 1$): For $n = 1$, if we evaluate both sides:

$$\text{LHS} = f(1) = F_1 = 1, \quad \text{RHS} = F_{1+2} - 1 = F_3 - 1 = 2 - 1 = 1.$$

Since $\text{LHS} = \text{RHS}$, the inequality holds true.

2. We prove the induction

$$F_1 + F_2 + F_3 + \cdots + F_k = F_{k+2} - 1.$$

$$F_1 + F_2 + \cdots + F_k + F_{k+1} = F_{(k+1)+2} - 1 = F_{k+3} - 1.$$

$$\begin{aligned} \sum_{i=1}^{k+1} F_i &= \left(\sum_{i=1}^k F_i \right) + F_{k+1} \\ &= (F_{k+2} - 1) + F_{k+1} \\ &= (F_{k+2} + F_{k+1}) - 1. \end{aligned}$$

We can rewrite F_{k+3} as, $F_{k+3} = F_{k+2} + F_{k+1}$. Substituting this in:

$$\sum_{i=1}^{k+1} F_i = F_{k+3} - 1.$$

Thus the equality holds true. □

Problem 3.5

Proof.

$$4^n + 7 \leq 5^n$$

1. Base Case ($n = 2$): For $n = 2$, we evaluate both sides:

$$\text{LHS} = 4^2 + 7 = 16 + 7 = 23, \quad \text{RHS} = 5^2 = 25.$$

Since $23 \leq 25$, the equality holds true.



2. We prove the induction

$$4^k + 7 \leq 5^k.$$

$$\begin{aligned} 4^{k+1} + 7 &= 4 \cdot 4^k + 7 \\ &< 5 \cdot 4^k + 35 \quad (\text{since } 4 \cdot 4^k < 5 \cdot 4^k \text{ and } 7 < 35) \\ &= 5(4^k + 7). \end{aligned}$$

$$5(4^k + 7) \leq 5 \cdot (5^k) = 5^{k+1}.$$

$$4^{k+1} + 7 \leq 5^{k+1}.$$

Thus, the inequality holds. □

Problem 3.6

A positive integer n is called a **perfect number** if it is equal to the sum of its proper positive divisors. Prove by induction that if $2^k - 1$ is a prime number (a **Mersenne prime**), then $n = 2^{k-1}(2^k - 1)$ is an even perfect number.

Proof. Let $p = 2^k - 1$ be a Mersenne prime. Since p is prime and $p > 1$, we must have $k \geq 2$, so $n = 2^{k-1}p$ is divisible by 2 and is therefore even.

The proper positive divisors of $n = 2^{k-1}p$ consist of:

1. Divisors that are powers of 2: $1, 2, 2^2, \dots, 2^{k-1}$.
2. Divisors multiplied by p (excluding n itself): $p, 2p, 2^2p, \dots, 2^{k-2}p$.

Thus, the sum of all proper positive divisors of n is:

$$S = \sum_{i=0}^{k-1} 2^i + p \sum_{i=0}^{k-2} 2^i.$$

We first prove by induction that for any integer $m \geq 0$:

$$g(m) = \sum_{i=0}^m 2^i = 2^{m+1} - 1.$$

1. Base Case ($m = 0$): For $m = 0$:

$$\text{LHS} = g(0) = 2^0 = 1, \quad \text{RHS} = 2^{0+1} - 1 = 2 - 1 = 1.$$

2. We prove the induction

$$\sum_{i=0}^{j+1} 2^i = \left(\sum_{i=0}^j 2^i \right) + 2^{j+1} = (2^{j+1} - 1) + 2^{j+1} = 2 \cdot 2^{j+1} - 1 = 2^{j+2} - 1.$$

Thus, the sum formula holds for all $m \geq 0$.



Using $g(k-1)$ and $g(k-2)$:

$$\sum_{i=0}^{k-1} 2^i = 2^k - 1 = p, \quad \sum_{i=0}^{k-2} 2^i = 2^{k-1} - 1.$$

$$S = p + p(2^{k-1} - 1) = p(1 + 2^{k-1} - 1) = p \cdot 2^{k-1} = n.$$

Since the sum of the proper positive divisors of n equals n , n is a perfect number. Since $k \geq 2$, n is an even perfect number. $\square$

Problem 3.7

Proof.

$$F_n < 2^n.$$

1. Base Cases ($n = 1$ and $n = 2$): For $n = 1$:

$$\text{LHS} = F_1 = 1, \quad \text{RHS} = 2^1 = 2.$$

Since $1 < 2$, the inequality holds true.

For $n = 2$:

$$\text{LHS} = F_2 = 1, \quad \text{RHS} = 2^2 = 4.$$

Since $1 < 4$, the inequality holds true.

2. We prove the induction Assume by strong induction that $F_i < 2^i$ holds for all $1 \leq i \leq k$, where $k \geq 2$. We show that $F_{k+1} < 2^{k+1}$.

By definition of the Fibonacci sequence:

$$F_{k+1} = F_k + F_{k-1}.$$

Applying the strong inductive hypothesis to F_k and F_{k-1} (since $1 \leq k-1 < k$):

$$F_k < 2^k \quad \text{and} \quad F_{k-1} < 2^{k-1}.$$

Substituting these into the expression for F_{k+1} :

$$F_{k+1} = F_k + F_{k-1} < 2^k + 2^{k-1}.$$

Since $2^{k-1} < 2^k$ for all $k \geq 2$:

$$2^k + 2^{k-1} < 2^k + 2^k = 2 \cdot 2^k = 2^{k+1}.$$

Therefore:

$$F_{k+1} < 2^{k+1}.$$

Thus, the inequality holds. $\square$